

Suraj Mishra

551-371-2172 | sm12377@nyu.edu | linkedin.com/in/msuraj001 | github.com/surajm20061998 | [Portfolio](#)

SUMMARY

Engineer working on empirical interpretability of safety-critical LLM failure modes. Recent work on Qwen3-14B and Qwen3-8B identifies a mechanistic explanation for long-context jailbreaks: refusal-mediating “Guardrail Heads” lose attention mass on harmful tokens as benign context grows, refusal halves with as little as 31 tokens of harmless padding, and canonical activation-steering fails to rescue, indicating the failure lives inside the attention mechanism rather than in the read-off layer. Broader interests include mechanistic interpretability of alignment failures, model organisms of misalignment, and adversarial robustness. Prior experience includes nearly 4 years shipping reliability-critical distributed systems.

EDUCATION

New York University

MS, Information Systems

New York City, NY

Sept. 2024 – May 2026

- **Coursework:** Deep Learning, Machine Learning, Building LLM Reasoners, Robo-Advisors & Systematic Trading, Operating Systems, DBMS

National Institute of Technology

B.Tech, Computer Science and Engineering

Bhopal, India

Jul. 2016 – Jun. 2020

EXPERIENCE

Member Technical Staff

Oracle

Jan 2022 – Aug 2024

Bangalore, India

- Designed fault-tolerant distributed backup/recovery infrastructure on OCI (Oracle 19c/23c) at petabyte scale under strict SLAs.
- Resolved throughput bottleneck via CPU/IO profiling and resource-aware parallel scheduling, improving throughput $\sim 80\%$.
- Built CI/CD pipelines (Jenkins, Terraform, Docker, Kubernetes) enabling weekly production releases.

Software Engineer

Incture Technologies

Sep 2020 – Dec 2021

Bangalore, India

- Scaled Spring Boot microservices for supply-chain analytics, sustaining $5\times$ traffic growth; cut API p99 latency $\sim 70\%$ via SQL profiling and connection-pool tuning.

PROJECTS

Attention Dilution in Safety-Tuned LLMs | *PyTorch, TransformerLens, circuit-tracer*

Feb 2026 – Apr 2026

- Code & full writeup (joint repository; my work in /Suraj).
- **Mechanistic finding:** long-context jailbreaks succeed because specific “Guardrail Heads” (identified via direct logit attribution onto $\hat{V}_{refusal}$) lose attention mass on the harmful tokens as benign context grows — a $6\times$ drop on Qwen3-14B and $44\times$ on Qwen3-8B for the top guardrail head between $N = 0$ and long N . Refusal collapse is *attentional*, not representational: the harmful-span residual cosine to $\hat{V}_{refusal}$ is flat across four orders of magnitude of N .
- **Jailbreak threshold:** on Qwen3-14B with the **distractor** bloat format, refusal halves at **31.5 bloat tokens** — a paragraph of harmless padding, not “long context.” Showed that bloat *format* matters as much as length: **distractor** dilutes attention $\sim 100\times$ more efficiently than **prefix** bloat on 14B, and the most-effective format differs across models in the same family.
- **Negative result on canonical rescue:** naive activation steering at the read-off layer (adding $\alpha \cdot \hat{V}_{refusal}$ for α up to 16) *fails* to recover refusal at long N on both models, while leaving MMLU/GSM8K capability and benign-prompt refusal untouched. Path-patching localizes recoverable signal to L20 ($\sim 9.3\%$), well below the read-off at L36, and shows the signal is spread across many heads — the failure lives in the attention mechanism, not in any single direction or layer.
- **Method:** extracted $\hat{V}_{refusal}$ via difference-of-means (Arditi et al., NeurIPS 2024) on 256 paired AdvBench / length-matched-Alpaca prompts; identified 12 Guardrail Heads via per-head DLA onto $\hat{V}_{refusal}$; ran a 7-phase pipeline measuring H1 (head attention to harmful span), H2a/H2b (residual cosine at readout vs. harmful-span position, following Zhao et al. NeurIPS 2025), H3 (steering rescue), capability cost, format triage, and head-level path patching. Single-direction ablation on 100-prompt held-out set drives harmful refusal $86\% \rightarrow 0\%$ on 14B with zero collateral damage to harmless prompts.

- **Engineering:** Persistent CSVs and OOM-aware hooks so partial runs re-launch only failed cells; ran on 1×A100-80GB (14B) and A100-40GB (8B) with bfloat16 inference under `transformer_lens`; piloted Anthropic’s `circuit-tracer` for attribution graphs.

An Empirical Autopsy of Small-Model GRPO | *PyTorch, TRL, Weights & Biases*

Jan 2026 – Present

- Studying GRPO training dynamics on a 1.5B model on Countdown with a focus on *when* and *how* reasoning behavior changes during RL, rather than only final accuracy; pilot results across 4 seeded runs over 20–40 steps show high seed variance, with best dev accuracy ranging from 0.277–0.340 and test accuracy from 0.229–0.292 at 20 steps, while one seed extended to 40 steps reached 0.342 test accuracy.
- Designed Phase-1 “autopsy” instrumentation that records fixed held-out probe rollouts at regular checkpoints during GRPO training, enabling step-by-step inspection of how the same prompts evolve rather than only comparing the initial and final policy; mean reward on the fixed probe set trends upward across recorded checkpoints, confirming sensitivity to genuine policy improvement.
- Captured per-token artifacts at each autopsy step, including log-probs, entropy tensors, rollout text, periodic model snapshots, and scalar training history, enabling planned downstream analysis of entropy collapse, reward hacking, format-vs-reasoning gains, and token-bucket dynamics across structure, numbers, operators, and text.
- Built a one-command autopsy suite using `run_autopsy_suite.sh` covering pilot runs across seeds, full GRPO runs, and off-policy contrast runs; integrated W&B artifact uploads so each run’s full output directory, including rollouts, tensors, and checkpoints, is preserved and comparable across experiments.

Compute-Optimal Scaling of Transformer LMs | *PyTorch, Weights & Biases, custom BPE*

Jan 2026 – Present

- Built an end-to-end LM training pipeline from scratch on TinyStoriesV2-GPT4, covering raw-text ingestion, custom BPE tokenization with vocab size 512, Transformer training, checkpointing, resumption, and scaling experiments.
- Engineered a parallel tokenization pipeline across 1, 2, 4, and 8 workers with throughput benchmarking, producing `.npy` token artifacts, vocab files, and merges managed as W&B dataset artifacts for reproducibility across runs.
- Trained Transformer LMs with context length 256, d_{model} 512, 8 layers, 8 heads, FFN size 1024, and 40k iterations, with online W&B logging, milestone checkpoint retention, and restartable runs directly from checkpoint artifacts.
- Ran a compute-optimal scaling sweep across 3 model sizes and 4 token budgets to characterize loss–compute trade-offs in a small-model language modeling regime.
- Currently extending the pipeline to mixture-of-experts (MoE) architectures to study routing dynamics and expert specialization under compute-optimal scaling.

TECHNICAL SKILLS

Interpretability & Alignment: TransformerLens, circuit-tracer, activation steering, difference-of-means probes, causal ablation & patching, attribution graphs, SAEs / MLP transcoders, LLM-as-judge evaluation

Training & RL: PyTorch, HuggingFace `transformers`, TRL, GRPO, SFT, custom BPE tokenization, Weights & Biases

Systems: CUDA, SLURM, Singularity, A100, bf16 mixed-precision, FlashAttention, KV-cache

Languages: Python, C++, Java, SQL, Bash

Infra & Tooling: Docker, Kubernetes, FastAPI, Jenkins, Terraform, uv